Particle Physics 3: Supersymmetry and Grand Unification

Lecture 1: Renormalization, Naturalness, and the Hierarchy Problem

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Chapter 1

Renormalization, Naturalness, and the Hierarchy Problem

Overview. The central ideas of this course grew out of puzzles about renormalization in the Standard Model. We begin, therefore, with the physical meaning of renormalization: eliminate degrees of freedom that are too small and too fast for the question at hand, then encode their effects in an effective description. A molecule supplies the prototype. Dimensional analysis then carries the same reasoning into scalar quantum field theory, where it exposes the quadratic sensitivity of a scalar mass, the hierarchy problem of the Higgs field, the protection enjoyed by fermions and gauge bosons, and the still more severe problem of vacuum energy.

1.1 Physics at the Scale of the Question

Most of the ideas to be developed this quarter originated in questions about renormalization—sometimes puzzles, sometimes useful observations—in the Standard Model. Renormalization combines two habits of thought. The first is to remove from a description structures at distances so short that they are irrelevant to the question being asked. The second is dimensional analysis, which often determines the short-distance answer before any difficult integral is evaluated.

This elimination of detail happens throughout physics. If we wish to understand a nucleus, we may begin with quarks and quantum chromodynamics. After calculating the masses and spins of the proton and neutron, together with their effective nuclear forces, we can forget their quark substructure for many purposes. Nuclear motion is slow enough that a nonrelativistic quantum mechanics of protons and neutrons is often the more useful description.

At the next level, atomic physics does not ordinarily need the internal constitution of a nucleus. Nuclear physics tells us which nuclei and isotopes exist and supplies such properties as mass and electric charge. We then begin again with a new set of effective elementary objects: nuclei and electrons. Quarks, protons, and neutrons have not ceased to exist; their effects have been compressed into the parameters carried by the nuclei.

The amount retained depends on the next question. An atomic calculation may need the nuclear mass because it enters the reduced mass, and it certainly needs the charge because that determines the Coulomb field. It usually does not need to know which quark carries which fraction of the nuclear momentum. There are roughly ninety elemental charges, with more nuclear species once isotopes are counted. That catalogue can be taken as input. Electrons, on the other hand, cannot yet be eliminated: their motion is exactly what makes atomic structure.

After solving the atomic problem, we may similarly regard atoms as the building blocks of molecules. Susskind's whimsical list of cookie atoms, coffee atoms, chocolate atoms, and ketchup atoms makes the serious point that a new description begins at every level. Once molecular properties and forces are known, molecules can be joined into larger Tinkertoy-like assemblies without reopening the electron problem at every step.

Each passage is a coarse-graining. The new account may be less exact in an absolute sense, but it is far more useful at its intended scale. Quantum field theory makes this issue unavoidable because a field has modes at every wavelength, including arbitrarily short ones. We therefore sum the effects of wavelengths too short to resolve and replace them by new effective parameters. That operation, together with dimensional analysis, is the physical content of renormalization.

1.2 A Molecule as the Prototype

Consider two atoms and ask for a simple description of their relative motion. Each atom contains a heavy nucleus and a cloud of electrons. Even the proton is roughly two thousand times heavier than an electron, and heavier nuclei increase the separation of scales. A useful picture is a pair of bowling balls surrounded by rapidly moving flies: the nuclei respond slowly while the electronic state rearranges quickly.

Let the nuclear positions and momenta be R_a and P_a, with common mass M, and let the electron coordinates and momenta be r_i and q_i, with mass m_e. Suppressing conventional factors such as $4\pi\epsilon_0$, the Hamiltonian has the schematic form

$$\begin{aligned} H = \frac{P_1^2}{2M} + \frac{P_2^2}{2M} + \frac{e^2}{R_{12}} + \sum_i \frac{q_i^2}{2m_e} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}} \\ - \sum_i \left(\frac{e^2}{|R_1 - r_i|} + \frac{e^2}{|R_2 - r_i|} \right). \end{aligned} \tag{1.1}$$

The first line contains the nuclear kinetic energies, their mutual Coulomb repulsion, and the electronic kinetic energy. The second line includes electron-electron repulsion and electron-nucleus attraction. For nuclei of general charges, the corresponding factors of Z_a are restored; the lecture keeps the charges schematic because the separation of time scales is the issue.

There are two characteristic motions. The nuclei are heavy and slow; the electrons are light and fast enough to look like a blur. In the first approximation, fix the nuclear positions and group all electron-dependent terms into an electronic Hamiltonian,

$$H_{\mathrm{el}}(R_1, R_2). \tag{1.2}$$

For each fixed nuclear configuration, solve

$$H_{\mathrm{el}}(R_1, R_2)\psi_0(r_1, \ldots, r_N; R_1, R_2) = E_{\mathrm{el}}(R_1, R_2)\psi_0. \tag{1.3}$$

We retain the lowest electronic eigenvalue. The electrons have not been ignored; all their rapid motion has been summarized by the function $E_{\mathrm{el}}(R_1, R_2)$.

The logic is adiabatic. During one rapid electronic adjustment, the nuclei have moved so little that their coordinates can be treated as parameters. After the electronic eigenproblem is solved, those same coordinates are allowed to move again. At each slowly changing configuration the electron cloud follows its instantaneous ground state. A more accurate treatment could retain excited electronic surfaces and transitions between them, but the leading surface already exhibits the renormalization idea.

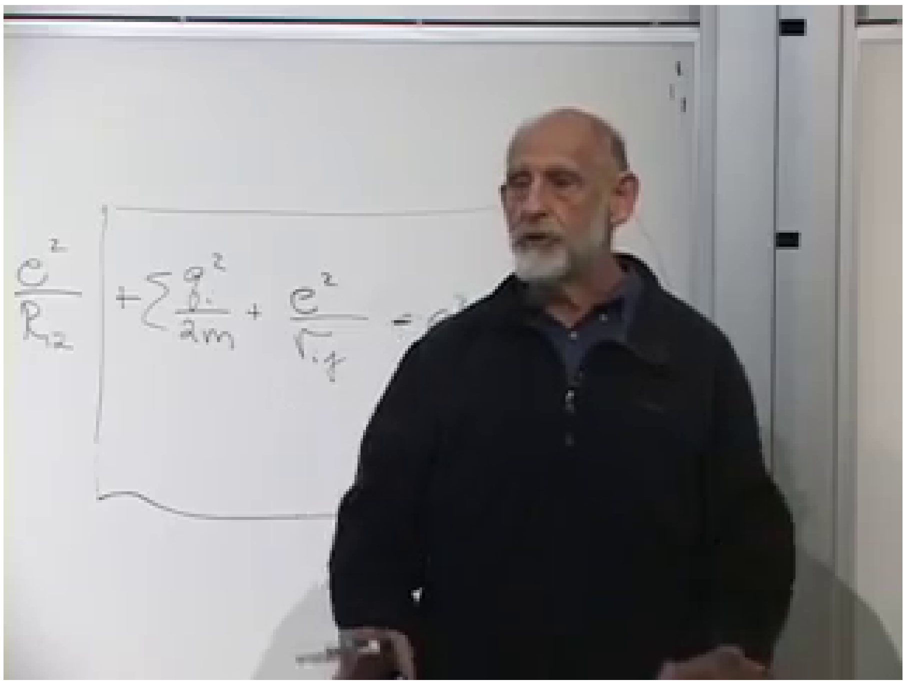

Figure 1.1: The molecular Hamiltonian remains on the board as the electronic ground-state problem is introduced. The typed form in Eq. (1.1) records the terms obscured in this frame.

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Question from the audience. Are the nuclear momenta simply being set to zero?[a]

Response. Only temporarily. In the leading approximation the nuclei are treated as nailed down because they move much more slowly than the electrons. Their momenta will be restored when we solve the effective nuclear problem, so this approximation is not a violation of the uncertainty principle.

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[a] Lecture timestamp: 00:15:39.

The electronic ground-state energy is now a contribution to the potential felt by the nuclei:

$$V_{\text{eff}}(R_1, R_2) = \frac{e^2}{R_{12}} + E_{\text{el}}(R_1, R_2). \tag{1.4}$$

Rotational and translational invariance reduce it, for two isolated atoms, to a function of their separation R_{12}. At very short distance the nuclear Coulomb repulsion dominates. At very large distance there is little interaction. Between these limits the electronic cloud can produce an attractive region and a minimum. The nuclear wavefunction then sits near that minimum and describes a bound molecule.

Notice the order of operations. We did not guess an empirical attraction and discard the electrons. We first solved their quantum problem in the fixed nuclear background. Only then did we erase their explicit coordinates and retain the resulting eigenvalue. The short-distance physics survives in the shape, depth, and curvature of V_{eff}. The location of its minimum sets the equilibrium separation, its curvature controls small vibrations, and its asymptotic form controls whether a bound state exists. One effective function carries all of that electronic information into the slower problem.

Thus a complicated many-electron Hamiltonian has become a two-body nuclear Hamiltonian with a new potential. We eliminated the fast degrees of freedom, but retained their complete low-energy effect. The potential has been renormalized. This familiar Born–Oppenheimer reasoning is not usually introduced under that name, yet it contains precisely the idea we need:

$$\begin{gathered}\text{fast microscopic dynamics}\\ \Downarrow\\ \text{effective parameters for slow dynamics}\end{gathered} \tag{1.5}$$

1.3 Natural Units and the Dimension of a Field

In quantum field theory, the modes to be eliminated may be described equivalently as short-distance, short-wavelength, high-frequency, or high-energy modes. Before drawing diagrams, let us fix the dimensional bookkeeping.

Ordinary mechanics uses independent units of length, time, and mass. In particle physics we set

$$\hbar = c = 1. \tag{1.6}$$

Then energy and mass are equivalent by $E = mc^2$, length and time are equivalent because $c = 1$, and $\hbar = 1$ makes momentum the inverse of length. Only one dimension remains:

$$[M] = [E] = [p] = L^{-1}, \qquad [x] = [t] = L. \tag{1.7}$$

This does not erase dimensions. It chooses conversion factors so that a time can be quoted as a length and a mass as an inverse length. Once one unit is specified, every other one follows. This is why a cutoff may be described either as a short distance δ or as a high energy $\Lambda = 1/\delta$: they are the same scale in two languages.

Take one real scalar field ϕ, with a schematically normalized Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi)(\partial^\mu \phi) - V(\phi), \qquad S = \int d^4x\, \mathcal{L}. \tag{1.8}$$

The Lagrangian density is local: it describes one spacetime point. Classical field equations follow by making the action stationary. In quantum theory the same action appears in the path-integral phase $\exp(iS)$. Either viewpoint makes the dimension of S decisive. With $\hbar$ restored, the phase is $\exp(iS/\hbar)$; after setting $\hbar = 1$, S itself must be dimensionless.

Question from the audience. Does the action have units of $\hbar$?[a]

Response. Yes. Since $\hbar$ has been set equal to one, the action is dimensionless in these units. The four-dimensional measure has dimension L^4, so the Lagrangian density must have dimension L^{-4}.

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[a]Lecture timestamp: 00:26:13.

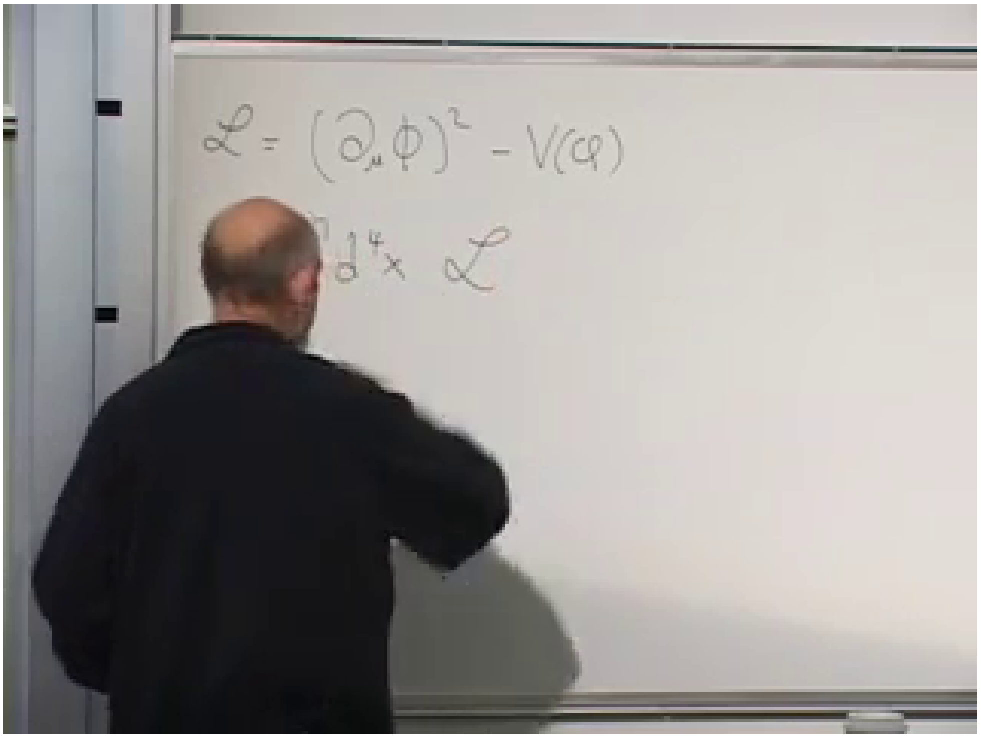

Figure 1.2: The local scalar Lagrangian density and its spacetime integral into the action.
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The kinetic term now fixes the dimension of the field:

$$[S] = 1, \qquad [\mathcal{L}] = L^{-4}, \tag{1.9}$$

$$\begin{aligned} [\partial_\mu] = L^{-1}, \qquad [(\partial\phi)^2] &= L^{-2}[\phi]^2 = L^{-4}, \\ [\phi] &= L^{-1}. \end{aligned} \tag{1.10}$$

This small calculation will determine the ultraviolet size of every diagram that follows.

Expand the potential as

$$V(\phi) = \frac{m^2}{2}\phi^2 + g\phi^3 + \lambda\phi^4 + \cdots . \tag{1.11}$$

Every term must have dimension L^{-4}. Since $[\phi^n] = L^{-n}$,

$$[m^2] = L^{-2}, \qquad [g] = L^{-1}, \qquad [\lambda] = 1. \tag{1.12}$$

The quartic coupling is dimensionless. Such couplings are especially important in renormalization because their scale dependence cannot be read off from a simple power of length.

Question from the audience. Does g have units of mass?[a]

Response. Yes. The factor ϕ^3 has dimension L^{-3}, while the potential density must have dimension L^{-4}. Therefore $[g] = L^{-1}$, which is a mass in natural units.

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[a] Lecture timestamp: 00:29:52.

1.4 Vertices and the Short-Distance Propagator

The Feynman rules are built from vertices and propagators. The term $g\phi^3$ gives a three-line vertex carrying a factor proportional to g; $\lambda\phi^4$ gives a four-line vertex carrying a factor proportional to λ. The quadratic term gives a local two-point insertion. It may be pictured as a particle entering, being absorbed, and being re-emitted, with a coefficient proportional to $m^2/2$. Numerical normalizations, signs, factors of i, and symmetry factors are deliberately suppressed because the lecture is extracting scale dependence rather than computing a cross section.

These drawings have an operational reading. A three-point vertex can absorb two quanta and emit one, or represent any crossing-related assignment of its three lines. A four-point vertex can describe two particles entering and two leaving. The quadratic cross is still a legitimate two-line vertex even though it is normally incorporated into the free propagator. We display it because every unresolved two-point process will shortly be recognized by comparison with it.

A propagator joins two spacetime points. Its line is not a claim that a quantum particle followed a straight classical path. It represents the amplitude to create a field excitation at x and remove or detect it at y:

$$\Delta(x-y) = \langle 0|\phi(y)\phi(x)|0\rangle. \tag{1.13}$$

Read the expression from the right. Acting on the vacuum with $\phi(x)$ creates a one-particle component at x. Acting with $\phi(y)$ on the bra removes the component at y. Their vacuum matrix element is the overlap amplitude between creation at the first point and detection at the second. The symbols x and y denote whole four-vectors, not merely two Cartesian coordinates.

States carry no physical dimension, while each field contributes L^{-1}. Therefore

$$[\Delta] = L^{-2}. \tag{1.14}$$

For a massless field there is no intrinsic length scale, so dimensional analysis forces the short-distance behavior

$$\Delta(x-y) \sim \frac{1}{(x-y)^2}. \tag{1.15}$$

The denominator is the Lorentz-invariant squared separation, with the usual prescription needed at its singularities.[1]

As y approaches x, the propagator diverges. This singularity of propagation over a vanishing separation is the source from which ultraviolet divergences in more elaborate diagrams are built.

1.5 How a Scalar Mass Is Renormalized

For a scalar field the dynamical parameter is m^2, as the relativistic dispersion relation already suggests:

$$E = \sqrt{\mathbf{p}^2 + m^2}. \tag{1.16}$$

Let δ be the shortest distance our effective description resolves. We shall not distinguish processes whose vertices are separated by less than δ. Any such unresolved process with one incoming and

[1] Editorial clarification: For power counting it is convenient to Wick rotate to Euclidean separation, where the magnitude is unambiguously proportional to $1/|x-y|^2$. This does not change the ultraviolet powers used below.

one outgoing scalar has the same observable two-point shape as a mass insertion and must therefore contribute to the effective mass.

Think of an accelerator as a microscope. If its finest resolution is δ, an absorption at one point followed by an emission a distance much smaller than δ away cannot be distinguished from a truly local event. We deliberately blur the vertex over that scale. Renormalization is the rule that transfers all such sub-resolution alternatives into the coefficient of the corresponding local operator.

1.5.1 The first quartic correction

Begin with a single ϕ^4 vertex. Two legs are the external particle; the other two join into a short loop emitted and reabsorbed at nearly the same point. Smearing the coincident propagator over the cutoff distance gives

$$\Delta(0) \;\longrightarrow\; \Delta(\delta) \sim \frac{1}{\delta^2}, \tag{1.17}$$

and hence

$$\delta m^2_{(1)} \sim \frac{\lambda}{\delta^2}. \tag{1.18}$$

If m_0^2 denotes the input or bare parameter, the low-resolution two-point coefficient has the schematic form

$$\frac{m^2_{\mathrm{eff}}}{2} \sim \frac{m_0^2}{2} + c_1 \frac{\lambda}{\delta^2}, \tag{1.19}$$

where c_1 contains the ignored numerical and convention-dependent factors. The laboratory mass measured in experiments that do not resolve distances below δ is this renormalized quantity, not m_0 by itself.

There are therefore two indistinguishable ways for the observed two-point event to occur. The original quadratic vertex can absorb and re-emit the particle, or the quartic interaction can do so while briefly emitting and reabsorbing a virtual scalar. Quantum mechanics adds the amplitudes. Once the loop lies below the resolution scale, its separate history is gone from the effective description, but its contribution to the measured coefficient is not.

1.5.2 The next quartic correction

Now connect two quartic vertices so that one external particle enters at x, one leaves from $x + \Delta$, and three internal propagators join the vertices. There are two powers of λ, while the propagators give Δ^{-6}. Quantum amplitudes must be summed over every unresolved place from which the outgoing particle can emerge, so the relative four-vector is integrated:

$$\delta m^2_{(2)} \sim \lambda^2 \int_{|\Delta| \gtrsim \delta} d^4\Delta \, \frac{1}{|\Delta|^6}. \tag{1.20}$$

Question from the audience. Does the small mark at the vertex mean there are only three particles there?[a]

Response. No. We are still using the quartic interaction. Four lines meet at each vertex; in this two-point diagram one line is external and three run between the two vertices. The second vertex again has four incident lines.

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[a] Lecture timestamp: 00:46:35.

The measure supplies four powers of length and the three propagators supply minus six. Since the ultraviolet cutoff is the only scale in the estimate,

$$\int_{|\Delta|\gtrsim\delta} \frac{d^4\Delta}{|\Delta|^6} \sim \frac{1}{\delta^2}, \qquad \delta m^2_{(2)} \sim \frac{\lambda^2}{\delta^2}. \tag{1.21}$$

Here Δ^μ is a four-component integration variable; one integrates over its time and three spatial components. The lowercase δ is the fixed cutoff. The lecture initially uses the same spoken name for both and later corrects the notation.

The same result is visible by writing the four-dimensional measure radially. Up to an angular constant,

$$\int_\delta \frac{d^4\Delta}{|\Delta|^6} \sim \int_\delta \frac{R^3\,dR}{R^6} = \int_\delta \frac{dR}{R^3} \sim \frac{1}{\delta^2}. \tag{1.22}$$

This is why the large-distance upper limit is unimportant for the ultraviolet estimate: the integral is dominated by its lower endpoint.

Higher diagrams repeat the pattern. A diagram with more quartic vertices adds a higher power of λ, but dimensional analysis still permits a quadratically divergent mass contribution:

$$\begin{aligned} m^2_{\text{eff}} = m_0^2 + \frac{1}{\delta^2}\left(c_1\lambda + c_2\lambda^2 + c_3\lambda^3 + \cdots\right) \\ + \text{weaker terms.} \end{aligned} \tag{1.23}$$

1.5.3 The cubic interaction and a logarithm

Two cubic vertices contain two propagators between nearby points. Their contribution scales as

$$\delta m_g^2 \sim g^2 \int_{|\Delta|\gtrsim\delta} \frac{d^4\Delta}{|\Delta|^4}. \tag{1.24}$$

Now the measure and denominator carry equal powers. The remaining dependence is logarithmic rather than a power:

$$\delta m_g^2 \sim c_g g^2 \log\left(\frac{1}{\mu\delta}\right). \tag{1.25}$$

The reference scale μ makes the logarithm dimensionless. This is the fully dimensional version of the lecture's schematic $g^2 \log\delta$. A logarithm varies much more slowly than any inverse power, so $1/\delta^2$ dominates as the cutoff becomes short.

Again the radial form makes the distinction transparent:

$$\int_\delta^{R_{\text{IR}}} \frac{d^4\Delta}{|\Delta|^4} \sim \int_\delta^{R_{\text{IR}}} \frac{R^3\,dR}{R^4} = \log\left(\frac{R_{\text{IR}}}{\delta}\right). \tag{1.26}$$

Equal powers in measure and denominator produce a logarithm. More powers in the denominator produce an inverse power of the cutoff. The lecture is not asking us to evaluate angular factors; it is teaching us to recognize the ultraviolet answer on sight.

Question from the audience. To which base is the logarithm taken?[a]

Response. Use the natural logarithm. Changing the base only multiplies it by a numerical constant, and that constant is absorbed into the coefficient whose factors of 2, π, and symmetry numbers have not been calculated here.

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[a] Lecture timestamp: 00:57:42.

Question from the audience. How can one take the logarithm of a length, and where do the units of the mass correction come from?[a]

Response. A logarithm always compares two scales; properly it is the logarithm of a dimensionless ratio such as $1/(\mu\delta)$. The coupling g has one power of mass, so g^2 already has exactly the mass-squared dimension required for the correction.

[a]Lecture timestamp: 00:58:32.

Question from the audience. A Feynman diagram gives an amplitude. Is its squared magnitude a probability?[a]

Response. Yes, when the amplitude is placed in a physical transition calculation with the required state and phase-space factors. Here its more immediate role is different: an unresolved two-point amplitude has the same local form as the mass term and therefore shifts the effective mass parameter.

[a]Lecture timestamp: 00:59:41.

1.6 Every Parameter Becomes Effective

The infinite family of mass diagrams does not disappear merely because λ is small. Perturbation theory orders them by powers of the coupling, so successive terms may become smaller, but there are still infinitely many. The same short-distance elimination also renormalizes the interactions themselves.

Question from the audience. Even for mass renormalization, are there not infinitely many diagrams?[a]

Response. There are. If λ is sufficiently small, the perturbative series may be useful because higher powers are successively suppressed. Nevertheless, the renormalized parameter is defined by the whole series, not by the first drawing alone.

[a]Lecture timestamp: 01:00:35.

Operationally, a local $\lambda\phi^4$ interaction is a two-particle-in, two-particle-out amplitude. If a more complicated four-point process occurs entirely inside an unresolved region, it is indistinguishable at low resolution from the original local vertex. Such diagrams are therefore lumped into an effective λ. Their leading scale dependence is typically logarithmic in four-dimensional ϕ^4 theory. The lecture does not calculate that beta function; it uses the example to make the general statement:

$$\begin{gathered}\text{parameters measured at low energy}\\ \neq\\ \text{parameters entered at the microscopic cutoff}\end{gathered} \tag{1.27}$$

The four-point example matters because it removes a possible misunderstanding. Renormalization is not a special repair applied only to a mass divergence. Every operator allowed in the Lagrangian can be reproduced by unresolved diagrams with the same external legs and symmetries. A two-point process renormalizes a quadratic coefficient, a four-point process renormalizes a quartic coefficient, and vacuum-to-vacuum processes will renormalize the constant term. The low-energy Lagrangian is the compact ledger in which all of those short-distance processes are recorded.

Question from the audience. Since Δ has units of length, is a logarithm of it not dimensionally strange? And did the board interchange uppercase and lowercase delta?[a]

Response. The logarithm really means a ratio to another scale and is therefore dimensionless. Yes, the notation was briefly interchanged: Δ^μ is the integration variable, while δ is the lower distance cutoff that remains in the answer.

[a]Lecture timestamp: 01:01:34.

This is the field-theory counterpart of the molecule. We solved for the unseen fast electrons and put their result into a potential. Here we sum unseen short-distance processes and put their result into masses, couplings, and every other allowed local coefficient. Everything is shifted.

Question from the audience. Could you repeat whether the measured quantity includes scales smaller than the experiment can see?[a]

Response. It does. Their individual fluctuations are not resolved, but their summed effect is present. All diagrams involving shorter scales are collected into the single effective coefficient that a low-energy experiment calls, for example, the physical mass squared.

[a]Lecture timestamp: 01:05:13.

Question from the audience. Must the input term m_0^2 then cancel the other contributions?[a]

Response. If the measured scalar mass is far smaller than the individual ultraviolet contributions, yes. The required cancellation is the fine-tuning problem to which we now turn.

[a]Lecture timestamp: 01:06:00.

1.7 The Higgs Hierarchy Problem

Imagine that the cutoff is the Planck length, so that the effective field theory is being asked to summarize fluctuations from laboratory scales down to the scale where quantum gravity must intervene. In particle-physics units,

$$\delta \sim M_{\mathrm{Pl}}^{-1} \sim 10^{-19}\,\mathrm{GeV}^{-1}, \qquad \frac{1}{\delta^2} \sim 10^{38}\,\mathrm{GeV}^2. \tag{1.28}$$

The precise use of the reduced or unreduced Planck scale is irrelevant to the enormous comparison.

The distance hierarchy is equally striking. The Planck length is about nineteen orders of magnitude below a proton radius and roughly sixteen or seventeen orders below the distances directly probed by the accelerators being discussed. Squaring that separation is what turns a large hierarchy of lengths into the roughly thirty-four-place cancellation in a scalar mass-squared parameter.

For the Higgs field, a scalar, the desired weak scale is of order a few hundred GeV, not 10^{19} GeV. Yet the terms in Eq. (1.23) can naturally be of Planck-scale size. Their signs have not been tracked: some may be positive and some negative, and the bare mass squared may also have either sign. But to leave an electroweak-scale remainder, those huge pieces must cancel to roughly thirty-four decimal places. Susskind first says seventeen digits, then corrects himself because the comparison is between squared masses. It is the squared ratio that measures this version of the tuning.

Question from the audience. Is λ smaller or larger than one?[a]

Response. It is assumed to be modestly smaller than one. Perturbation theory would become intractable if successive powers grew, and a pure scalar theory also cannot be extrapolated consistently with an arbitrarily large quartic coupling. But λ is not remotely as small as 10^{-38}; the lecture takes a percent-scale number for orientation.

[a]Lecture timestamp: 01:09:50.

Question from the audience. Even if $\lambda < 1$, do not many terms remain important because each carries the large cutoff factor?[a]

Response. Yes. Higher powers eventually suppress the series, but a large collection of terms can contribute before that happens. They and the bare term would all have to conspire to leave the extraordinarily small physical answer.

[a]Lecture timestamp: 01:10:43.

Recall why the Higgs mass parameter is known to be weak-scale. In the lecture's convention, write the symmetry-breaking potential as

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4. \tag{1.29}$$

The negative quadratic coefficient makes the origin unstable, while the positive quartic term stabilizes the potential. Differentiating,

$$\frac{dV}{d\phi} = -\mu^2\phi + \lambda\phi^3 = \phi(-\mu^2 + \lambda\phi^2), \tag{1.30}$$

gives the nonzero minimum

$$\phi_{\rm min}^2 = f^2 = \frac{\mu^2}{\lambda}. \tag{1.31}$$

The board writes the quartic schematically as $\lambda\phi^4$ but quotes Eq. (1.31); inserting the conventional factor $1/4$ makes those two lines consistent without changing the argument.

The sign of $-\mu^2$ is important but does not rescue naturalness. It is the negative curvature at the origin that drives spontaneous symmetry breaking. Renormalization shifts that curvature by large positive and negative pieces. The observed broken vacuum requires their sum not merely to be small, but to have the particular small negative value that places the minimum at the weak scale.

The W- and Z-boson masses determine f to be of order 200 GeV in the rough normalization used in the lecture. If λ is neither enormous nor fantastically tiny, μ must be of the same general order. The observed weak-scale coefficient is therefore the residue of contributions that may each be tens of orders of magnitude larger.

This is called the Higgs fine-tuning problem, the hierarchy problem, or the gauge-hierarchy problem. The hierarchy is the immense ratio between the ordinary weak scale and a possible unification or Planck scale. A small number by itself need not be paradoxical. What is disturbing here is a small number assembled from large, apparently unrelated pieces that must cancel with extreme precision. The problem was recognized in the late 1970s and early 1980s, and the search for a mechanism behind that cancellation motivates supersymmetry and several competing ideas.

The infinite perturbative series sharpens rather than removes the puzzle. A percent-scale λ makes sufficiently high powers small, but many terms can occur before that suppression wins. It is not enough for one loop to cancel the bare mass accidentally. The effective weak scale must remain stable after the rest of the short-distance processes are included. A structural relation among the terms would be an explanation; an unrelated digit-by-digit cancellation is only a tuning.

1.8 Why a Fermion Mass Behaves Better

The other Standard Model particles do not require independent tunings of the same kind. For a Dirac fermion, the mass term is

$$\mathcal{L}_m = -m\bar{\psi}\psi = -m\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right). \tag{1.32}$$

It contains the mass, not its square, and it couples left and right chiralities. A local mass insertion therefore changes L into R, or R into L.

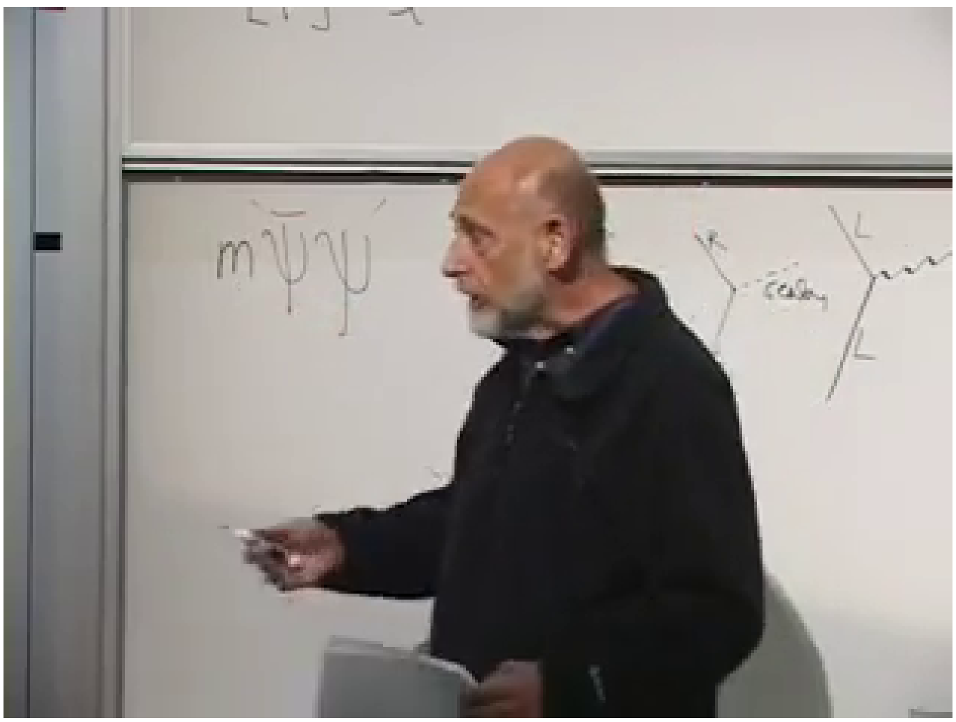

Figure 1.3: The fermion mass term and the board's chirality-changing line sketches. The diagrams are reconstructed algebraically below because part of the chalk work is occluded.

Lecture frame: 01:23:36.

The spoken lecture often says handedness or helicity. The exact protection is most cleanly stated in terms of chirality. For an ultrarelativistic particle the helicity picture supplies the intended intuition, but chirality is the quantity carried by the left- and right-handed fields in Eq. (1.32).

Gauge interactions preserve chirality:

$$L \xrightarrow{\gamma} L, \qquad R \xrightarrow{\gamma} R. \tag{1.33}$$

A Yukawa interaction with a scalar connects opposite chiralities:

$$L \xleftrightarrow{\phi} R. \tag{1.34}$$

Suppose first that the electron starts with $m_0 = 0$. Emitting and reabsorbing any number of photons cannot produce a mass, because every gauge vertex preserves chirality. A sequence that starts left remains left. Thus pure quantum electrodynamics consistently permits a massless electron: its electromagnetic self-energy cannot create an additive mass from zero.

In the line picture, start with a left-chiral electron. At the first photon vertex it is still left-chiral; after propagation and reabsorption of the photon it is still left-chiral. Repeating the construction only inserts more chirality-preserving vertices. The external two-point function can be renormalized, but it cannot acquire the left-right structure that defines a mass.

Could scalar emission do it? One scalar vertex changes L to R, but an emitted scalar must be reabsorbed. A two-point self-energy graph therefore contains an even number of scalar vertices. The second flip reverses the first, so the complete graph again preserves the external chirality. No matter how complicated the internal graph becomes, emission and reabsorption pair the vertices and prevent an additive L-to-R term from appearing when the starting mass is zero.

This is a parity argument on the number of flips. One scalar vertex flips chirality once; closing every internal scalar line requires a second endpoint and hence a second flip. Four scalar vertices flip four times, and so on. An even number returns the fermion to its original chirality. The conclusion is not restricted to the simplest loop drawn on the board.

A nonzero mass correction becomes possible only after the diagram already contains a mass insertion. For example, photon emission and reabsorption can surround one $m_0\bar{\psi}\psi$ insertion. The gauge vertices contribute two powers of the electric charge, but the sole chirality flip contributes the original mass:

$$\begin{aligned} m_e &= m_0 + c_1 e^2 m_0 + c_2 e^4 m_0 + \cdots \\ &= m_0 \left(1 + c_1 e^2 + c_2 e^4 + \cdots\right). \end{aligned} \tag{1.35}$$

If $m_0 = 0$, the result remains zero. If m_0 is small, every correction is proportional to that small number. We may still ask why the electron is light relative to a fundamental scale, but we are no longer subtracting enormous unrelated numbers to obtain it. In this sense a small fermion mass is technically natural.

The mass insertion makes the allowed sequence explicit. Begin with L. Photon emission preserves L; the insertion m_0 changes L to R; photon reabsorption preserves R. The diagram now has the external chirality of a mass term, but its value necessarily contains the factor m_0. This is the crucial contrast with Eq. (1.18), whose scalar correction remains nonzero even when the starting scalar mass is set to zero.

Gauge bosons have an analogous protection: gauge symmetry forbids an independent mass term, so a gauge boson that begins massless remains massless to all perturbative orders. In the Standard Model, the masses of fermions and the weak gauge bosons arise in response to the Higgs expectation value. Control the Higgs scale, and the remaining light masses do not require separate copies of the same catastrophic tuning.

This identifies the question supersymmetry is meant to address: can a theory prevent the Higgs scalar from receiving an enormous additive mass correction? The answer is yes in suitably supersymmetric settings, because bosonic and fermionic contributions are related rather than independent. The detailed mechanism begins in the next lecture.

1.9 The Other Fine-Tuning: Vacuum Energy

There is one still more severe renormalization problem, but it becomes observable only when gravity is included. In nongravitational physics, adding a constant to every energy changes no experiment. The electron mass, for example, is not an absolute energy; it is the extra energy required to add an electron to the vacuum:

$$m_e = E(\text{vacuum} + \text{one electron}) - E(\text{vacuum}) \tag{1.36}$$

in the electron rest frame. If the vacuum and every excited state shift together, their differences remain unchanged.

Nevertheless, vacuum-to-vacuum diagrams do shift the vacuum energy. In $\lambda\phi^4$ theory, join the four legs of one quartic vertex into two short loops. Each coincident propagator supplies $1/\delta^2$, so

$$\delta\rho_{\text{vac}} \sim c_{\text{vac}} \frac{\lambda}{\delta^4}. \tag{1.37}$$

More complicated vacuum diagrams carry different coupling factors and coefficients, but dimensional analysis typically leaves the same quartic cutoff sensitivity for an energy density.

The absence of external lines is exactly what identifies the correction. Two-point graphs compare a one-particle state with the vacuum and alter a mass; four-point graphs alter scattering; a graph with no incoming or outgoing particles changes the vacuum-to-vacuum amplitude itself. One quartic vertex supplies λ, and its two short loops supply δ^{-2} apiece. Dimensional analysis then requires the δ^{-4} energy-density scale.

Without gravity one may choose the additive zero of energy and ask who cares. General relativity removes that freedom: stress-energy sources spacetime curvature, so a uniform vacuum energy makes empty space gravitate. It appears as a cosmological constant and is connected observationally with dark energy. A tiny measured vacuum energy must then survive contributions of the form δ^{-4}.

This is why the question changes when gravity is turned on. In ordinary quantum mechanics, shifting every level by the same constant leaves all transition frequencies untouched. In gravity, the constant is not inert: the vacuum stress tensor is proportional to the metric and curves spacetime. What had been an arbitrary zero of energy becomes a source in Einstein's equation.

Thus there are at least two extreme naturalness puzzles. The lecture quotes roughly 123 decimal places for the vacuum-energy tuning and roughly 30–35 for the Higgs mass tuning. The exact counts depend on the ultraviolet scale and conventions, but the contrast is not in doubt. Supersymmetry is a potential response to the Higgs problem discussed here; once supersymmetry is broken, it does not by itself explain the observed cosmological constant.[2]

[2] Editorial clarification: This is also why an exactly supersymmetric boson–fermion cancellation is not a phenomenological solution to the observed vacuum energy: supersymmetry must be broken in our world.

1.10 Closing Classroom Questions

Question from the audience. The Higgs correction depends on the cutoff δ, but the cutoff seems arbitrary. Could it be the Planck length, ten times the Planck length, or something else? Does that make the fine-tuning arbitrary?[a]

Response. The numerical contribution depends on the shortest scale to which the effective theory is trusted. That dependence is physical bookkeeping, not an escape. Modes down to any scale where the Standard Model still makes sense are modes whose effects must be included. Choose δ at the distance where new physics is expected—perhaps the Planck scale, perhaps a unification scale near 10^{16} GeV. The contribution accumulated down to that scale is already a lower bound on the problem. Extending the same theory to still shorter distances only makes the quadratic sensitivity larger. We are obligated to explain what cancels the contribution from every scale between the laboratory and that ultraviolet boundary.

[a] Lecture timestamp: 01:33:50.

Question from the audience. What defines a scalar particle, and what is an example?[a]

Response. The Higgs is the intended example. A scalar does not transform into different components under a spatial rotation; equivalently, its particle quanta have spin zero. A photon is described by a vector field, and the gravitational field by a tensor. At the time of this 2010 lecture, no elementary spin-zero particle had yet been detected.[b]

[a] Lecture timestamp: 01:37:26.

[b] Editorial clarification: The Higgs boson was discovered at CERN in 2012 and its measured properties are consistent with spin zero.

Question from the audience. Is the Z particle also a gauge boson?[a]

Response. Yes. The Z, W^+, and W^- are spin-one gauge bosons, as is the photon. Their field equations and interactions are non-Abelian or electroweak generalizations of the Maxwell-like gauge description.

[a] Lecture timestamp: 01:38:31.

Question from the audience. Why does the bosonic ground-state contribution not cancel the fermionic one in the vacuum energy?[a]

Response. Fermion loops and corresponding boson loops can indeed enter with opposite signs. Cancellation is not automatic, however: the particle masses, multiplicities, and couplings must match. Even a small mismatch leaves a large cutoff-sensitive residue, and the infinite families of diagrams must match order by order. Exact supersymmetry enforces just such boson–fermion relations. Generic nonsupersymmetric particle content does not, and broken supersymmetry leaves a residual vacuum-energy problem.

[a] Lecture timestamp: 01:39:10.

1.11 Perspective

The route through the lecture is deliberate. A molecular calculation first shows that eliminating fast variables is neither neglect nor magic: their ground-state energy becomes a potential for the slow variables. Quantum field theory repeats that operation across wavelengths. Dimensional analysis then reveals which effective parameters are sensitive to the cutoff. Scalar masses receive additive terms proportional to δ^{-2}; a small Higgs scale therefore appears unnaturally delicate. Fermion masses are multiplicatively renormalized because chirality protects the zero-mass limit, and gauge symmetry similarly protects massless gauge bosons. Vacuum energy is more sensitive still, scaling as δ^{-4}, and gravity makes its absolute value observable.

Supersymmetry enters the course at exactly this point, not as ornamental new mathematics but as an attempt to make the scalar sector behave more like the protected sectors. The next task is to understand what symmetry can relate bosons and fermions strongly enough to organize those cancellations.